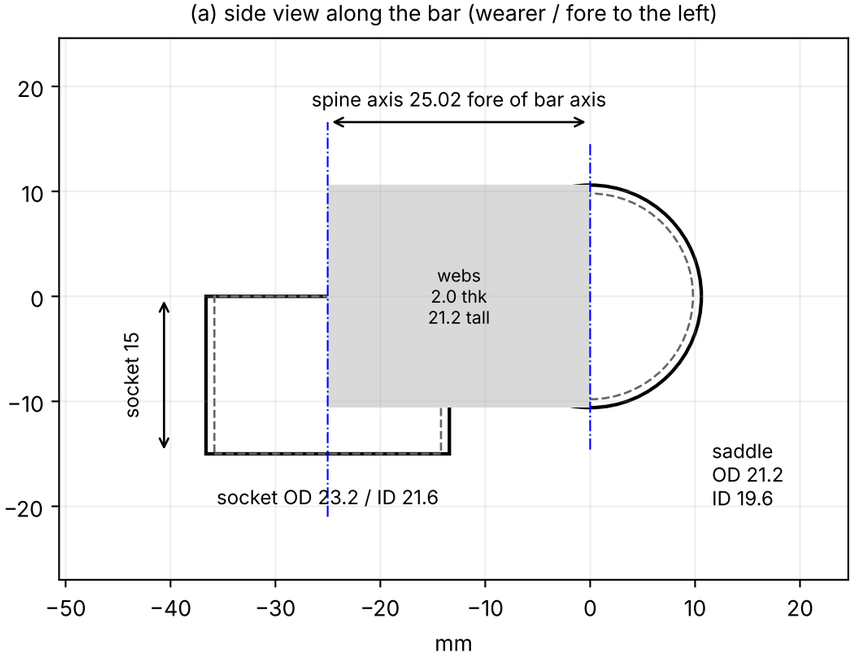
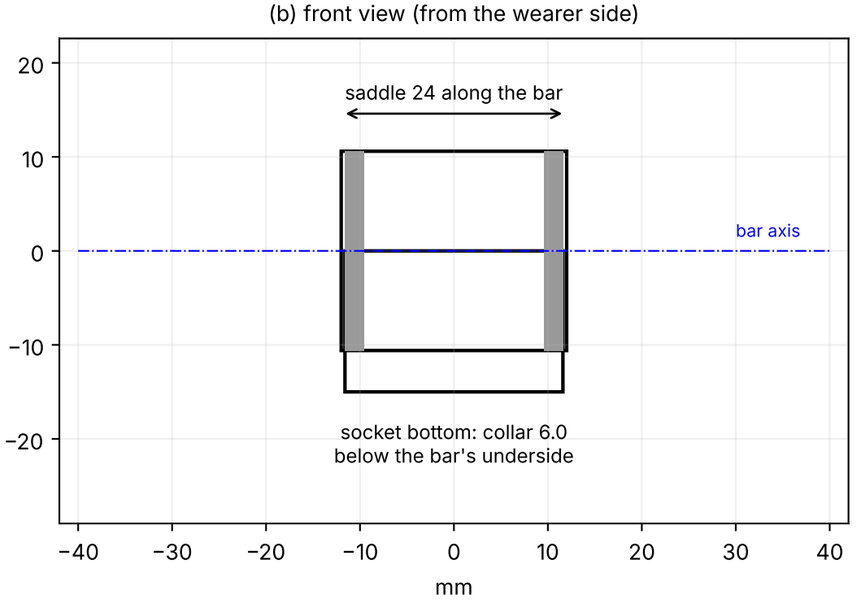
H1 crown spine clamp (BL-054, from huntkit/gendesign/crown_spine_clamp.py)






```
SPINE CLAMP, 7075-T6, 1 per frame (14.2 g)

Saddle: tube OD 21.2, ID 19.6, 24 long
   on the 18 mm top bar, 0.8 bondline
Socket: tube OD 23.2, ID 21.6, 15 deep
   on the 20 mm spine, 0.8 bondline,
   top of socket level with the bar axis
Spine axis 25.02 fore of the bar axis,
   parallel to the crown legs
Webs: 2 plates 2.0 thick x 21.2 tall,
   outer faces 23.2 apart, saddle to socket
Wall 0.8 everywhere; break edges 0.2
Finish: hard anodise (no bare Al on carbon)
Bond: MMA + 0.8 mm glass beads (BUILD_H1 13)

Margins (crown_spine_clamp.size_clamp):
  socket avg 2.422, peak 1.4
  saddle avg 5.262, peak 1.153
  saddle peak + shear 0.812 -> T6(c) coupon
  webs yield 3.01

Units mm. Design intent from code constants;
machinist confirms tolerances (bores +0.1/-0).
```